

CA-1 OPEN SOURCE FIREWALLS ASSIGNMENT

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Assessment Title:	CA1 - Open Source FW Assignment

Declaration

☑ I confirm that I have read and understood the University's Academic Integrity Policy.

[redacted]

☑ I have completed the TU Dublin AI Driving License theory test.

[redacted]

☑ I declare that this assessment submission is my own work.

☑ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

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Network Topology

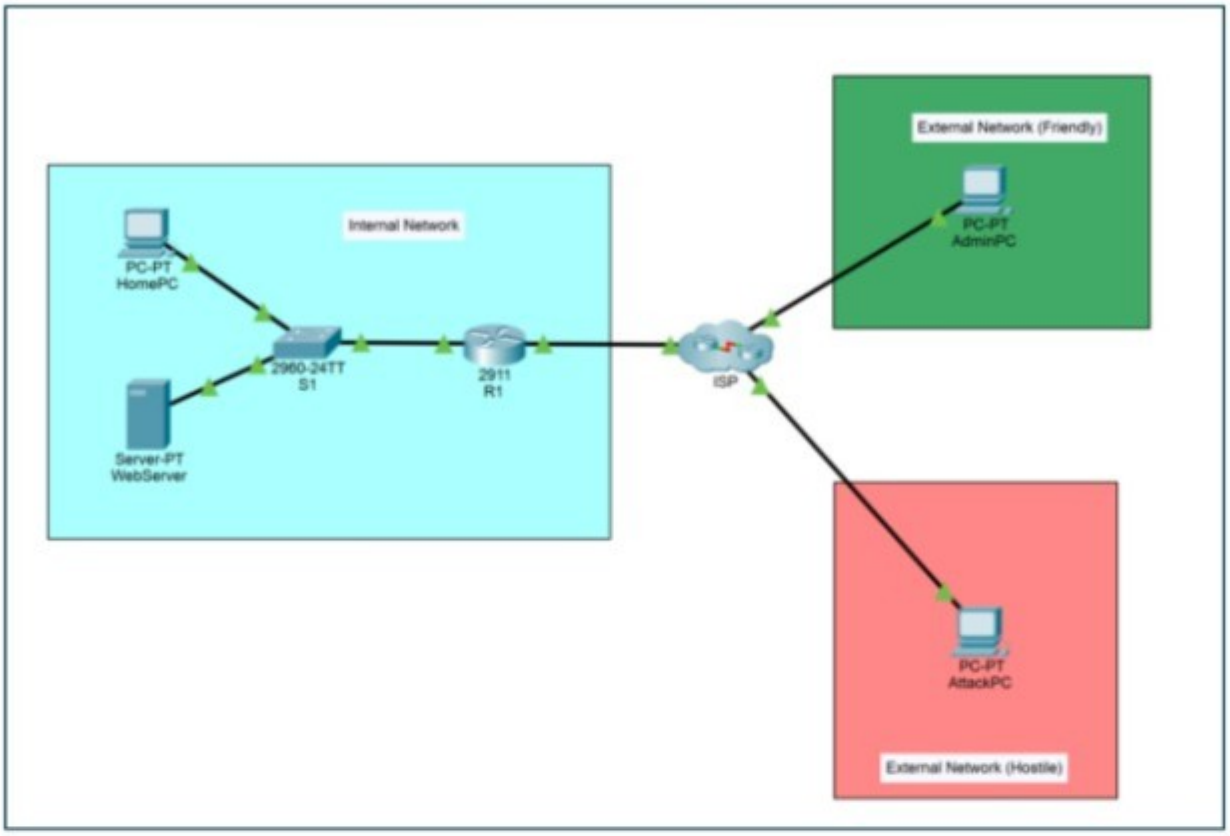


Figure 1: Network Topology

Network Setup

Kali Linux Attacker PC

- NIC 1: WAN_NET
- IP: 192.168.20.10/24
- Gateway: 192.168.20.1

Ubuntu Desktop Admin PC

- NIC 1: WAN_NET
- IP: 192.168.20.20/24
- Gateway: 192.168.20.1

Firewall (OPNsense / Smoothwall / VyOS)

- NIC 1: LAN_NET
- IP: 192.168.10.1/24
- NIC 2: WAN_NET
- IP: 192.168.20.1/24

Ubuntu Web Server

- NIC 1: LAN_NET
- IP: 192.168.10.10/24
- Gateway: 192.168.10.1

Ubuntu Desktop Home PC

- NIC 1: LAN_NET
- IP: 192.168.10.20/24
- Gateway: 192.168.10.1

Credentials Setup

Ubuntu Web Server

- Login: webserver
- Password: Pass1234

Ubuntu Desktop Home PC

- Login: homepc
- Password: Pass1234

Kali Linux Attacker PC

- Login: attackerpc
- Password: Pass1234

Ubuntu Desktop Admin PC

- Login: adminpc
- Password: Pass1234

Firewalls

OPNsense

- Login: root
- Password: o46JND)1u

Smoothwall

CLI

- Login: root
- Password: Pass1234

GUI (Web Interface)

- Website: https://192.168.10.1:441
- Login: admin
- Password: Pass1234

VyOS

- Login: vyos
- Password: vyos

Attacks

1. Reconnaissance (Passive + Active)
2. Port Scanning
3. Vulnerability Scanning
4. Brute Force / Credential Attacks
5. DoS / Resource Exhaustion (Rate-based, not destructive)

Desired Network Behaviour

Across all firewall implementations (OPNsense, Smoothwall, and VyOS), the expected and intended network behaviour for the test environment is as follows:

1. The Web Server can serve a web page to the WAN (Admin PC and Attacker PC) and the LAN (Home PC).
2. SSH connection to the Web Server accessible from Admin PC (that is on WAN) and not accessible to both Home PC (which is on LAN) and Attacker PC (which is on WAN).
3. Outbound traffic from the Web Server and Home PC allowed.
4. The OPNsense GUI and SSH are accessible to both the Home PC (on LAN) and the Admin PC (on WAN), but not to the Web Server (on LAN) or the Attacker PC (on LAN).

OPNsense – Mark Diadyk [redacted]

Introduction

OPNsense is a modern, FreeBSD-based open-source firewall and routing platform with a web GUI, stateful packet filtering (PF), NAT, and integrated IDS/IPS (Suricata).

Configuration

First boot

I have installed the OPNsense VM on my PC in VirtualBox using the .iso file from the official website. During the first boot into the live environment (in the CLI), I had to log in with the username "installer" and the password "opnsense" to begin the full installation. After that, I had to go through the standard installation steps: keymapping, selecting the target disk, and partitioning. I had to pick which interfaces were LAN and WAN. Then I set the interface IP addresses. I went to Home PC VM and opened OPNsense GUI in the web browser. By default, the username is "root" and the password is "opnsense". After logging in, I have changed the password to "o46JND)1u" for better security.

Firewall rules

I have set the firewall rules to achieve the desired network behaviour outlined in the upper section of this document.

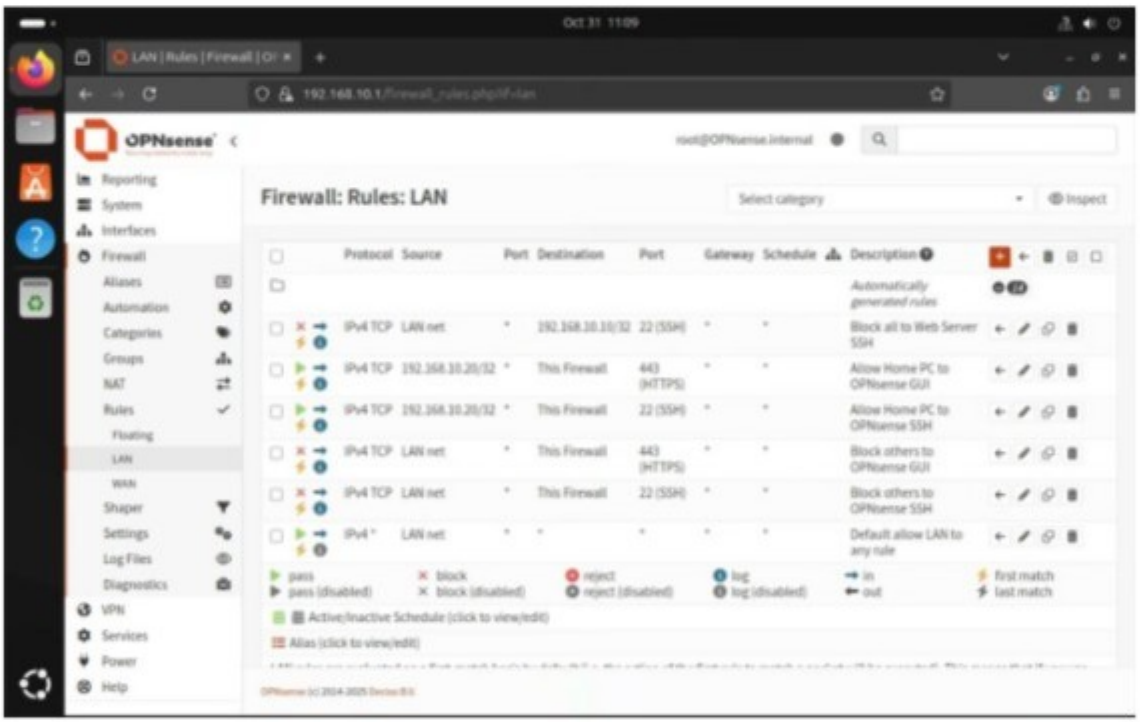


Figure 2: LAN Firewall Rules

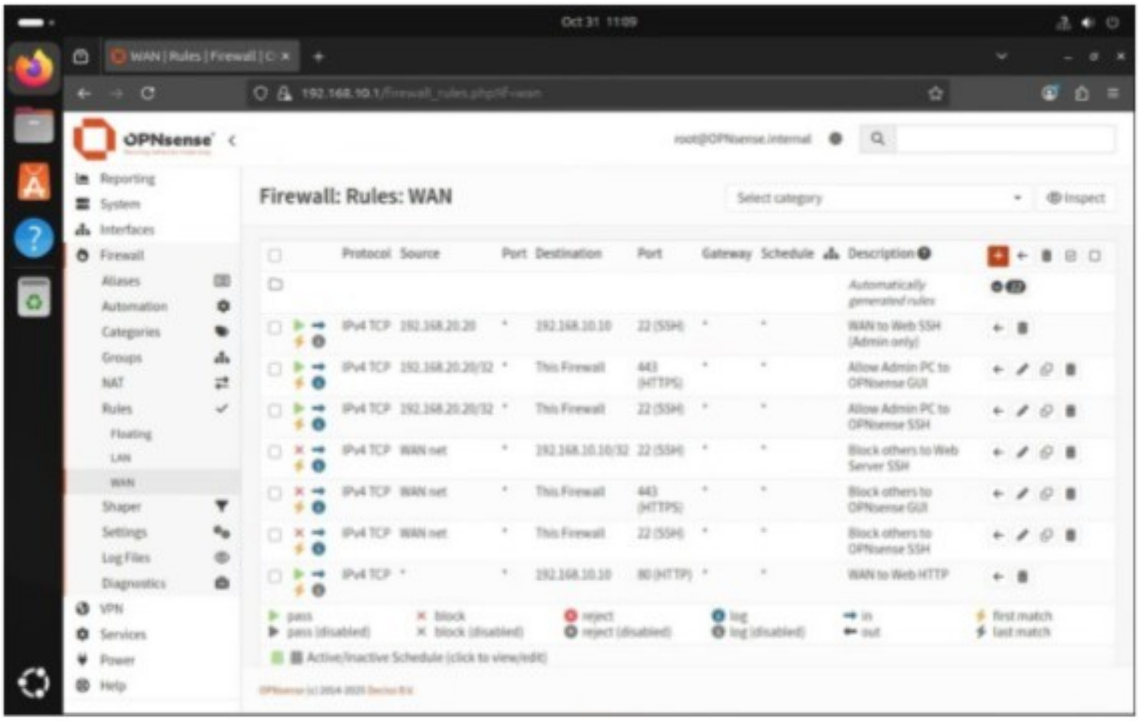


Figure 3: WAN Firewall Rules

NAT port forwarding

I have set up NAT port forwarding to make LAN services (web server HTTP and SSH) available to the WAN network.

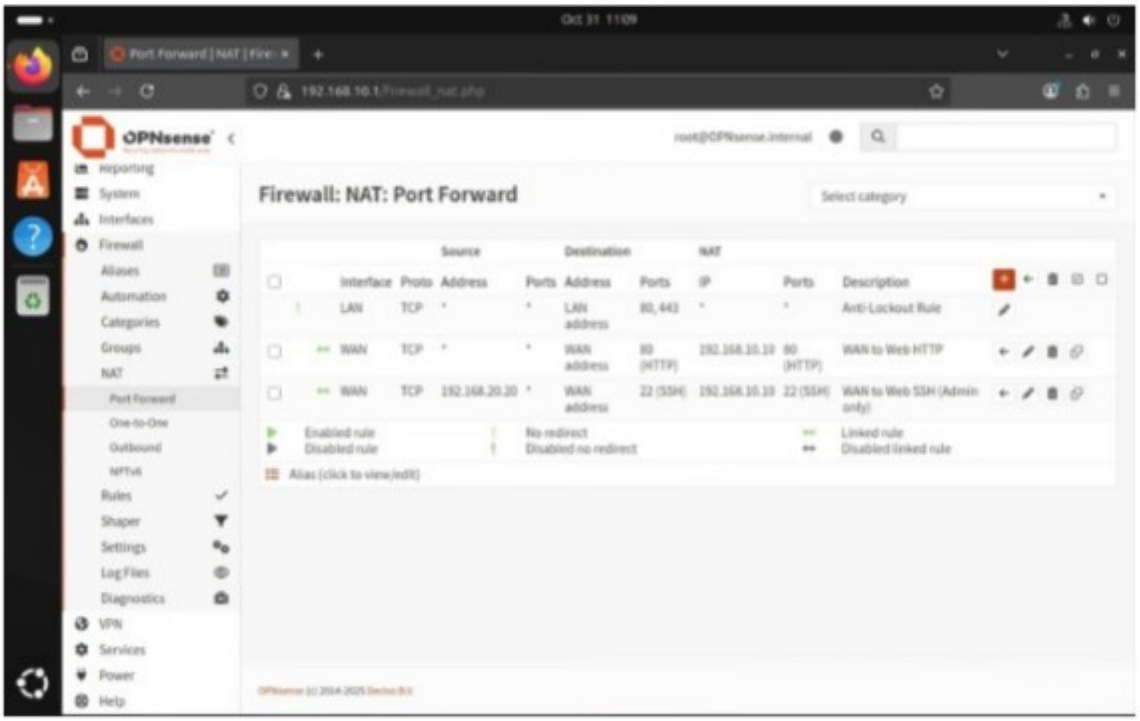


Figure 4: Firewall NAT Port Forwarding

Suricata intrusion detection

I have enabled Suricata intrusion detection on the WAN interface. Downloaded and enabled rulesets: ET open/botcc, ET open/emerging-dos, ET open/emerging-scan, ET open/emerging-web_client, ET open/emerging-web_server. Enabled select rules from those rulesets.

Intrusion detection helps detect ongoing attacks by matching the patterns of behaviour to the traffic.

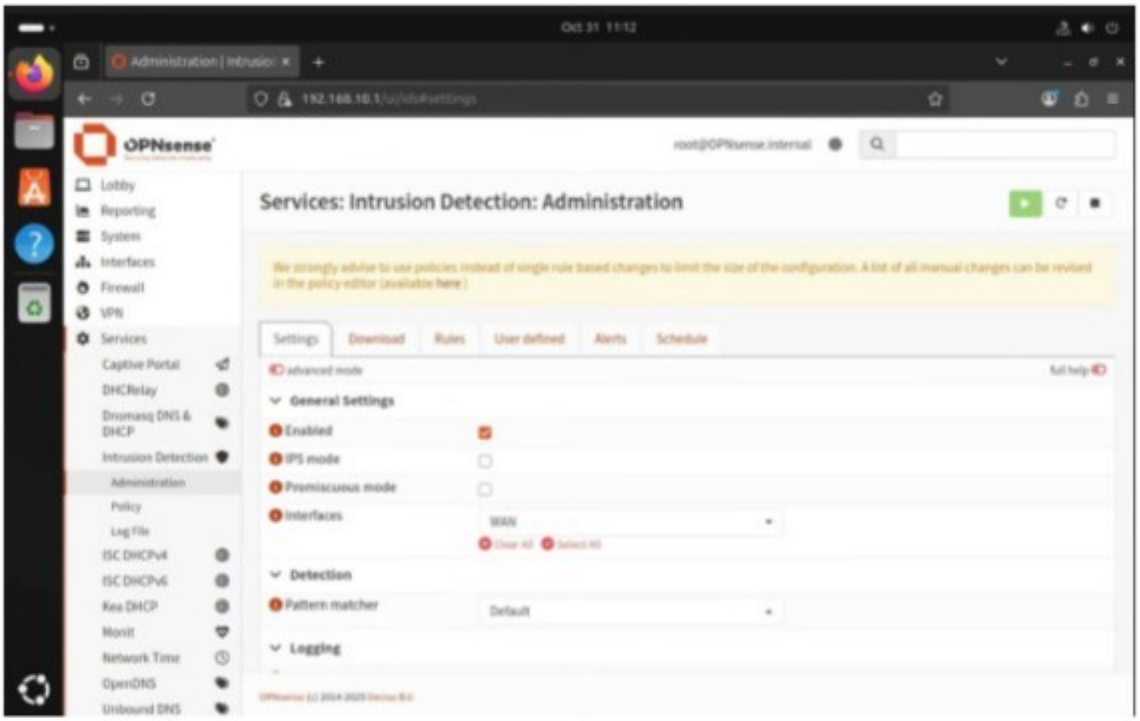


Figure 5: Intrusion Detection Settings

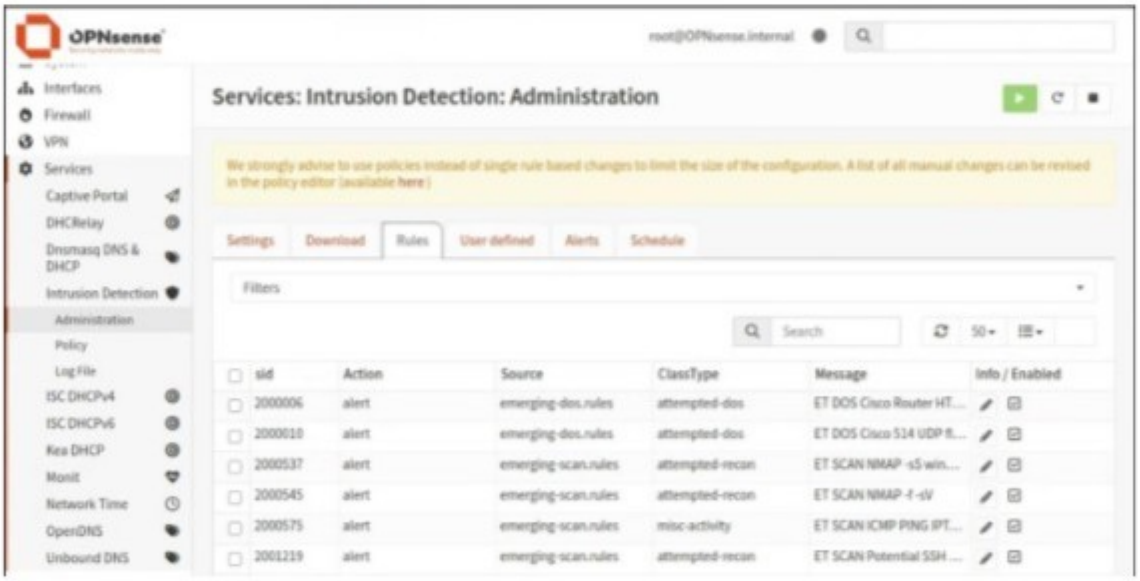


Figure 6: Intrusion Detection Rules Part 1

sid	Action	Source	ClassType	Message	Info / Enabled
2003634	alert	emerging-scan.rules	attempted-admin	ET SCAN Suspicious Us...	
2009159	alert	emerging-scan.rules	attempted-recon	ET SCAN Toata Scanner ...	
2009358	alert	emerging-scan.rules	web-application-attack	ET SCAN Nmap Scriptin...	
2009359	alert	emerging-scan.rules	web-application-attack	ET SCAN Nmap Scriptin...	
2009582	alert	emerging-scan.rules	attempted-recon	ET SCAN NMAP -sS win...	
2009583	alert	emerging-scan.rules	attempted-recon	ET SCAN NMAP -sS win...	
2009584	alert	emerging-scan.rules	attempted-recon	ET SCAN NMAP -sS win...	
2010088	alert	emerging-scan.rules	attempted-recon	ET SCAN Suspicious Us...	
2010089	alert	emerging-scan.rules	attempted-recon	ET SCAN Suspicious Us...	
2010371	alert	emerging-scan.rules	attempted-recon	ET SCAN Amap TCP Ser...	

Figure 7: Intrusion Detection Rules Part 2

sid	Action	Source	ClassType	Message	Info / Enabled
2011497	alert	emerging-scan.rules	attempted-recon	ET SCAN Hydra User-Ag...	
2011822	alert	emerging-dos.rules	denial-of-service	ET DOS User-Agent use...	
2011823	alert	emerging-dos.rules	denial-of-service	ET DOS User-Agent use...	
2011824	alert	emerging-dos.rules	denial-of-service	ET DOS User-Agent use...	
2014996	alert	emerging-dos.rules	attempted-dos	ET DOS Microsoft Wind...	
2016033	alert	emerging-scan.rules	web-application-attack	ET SCAN Simple Slowlo...	
2017722	alert	emerging-dos.rules	attempted-dos	ET DOS Trojan.BlackRe...	
2018782	alert	emerging-scan.rules	attempted-recon	ET SCAN Internet Scann...	
2019347	alert	emerging-dos.rules	attempted-dos	ET DOS HTTP GET AAAA...	
2019404	alert	emerging-dos.rules	attempted-dos	ET DOS Potential Tsuna...	

Figure 8: Intrusion Detection Rules Part 3

2034093	alert	emerging-dos.rules	denial-of-service	ET DOS Possible Micros...	
2034095	alert	emerging-dos.rules	denial-of-service	ET DOS Possible Apach...	
2034096	alert	emerging-dos.rules	denial-of-service	ET DOS Possible Apach...	
2100469	alert	emerging-scan.rules	attempted-recon	GPL SCAN PING NMAP	
2100628	alert	emerging-scan.rules	attempted-recon	GPL SCAN nmap TCP	

Figure 9: Intrusion Detection Rules Part 4

Attacks

Reconnaissance

I run “ping -c 4 192.168.10.10” command from the Attacker PC (WAN) towards the Web Server (LAN) to confirm that OPNsense enforces a default-deny rule against unsolicited WAN traffic.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ ping -c 4 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.

--- 192.168.10.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3057ms
```

Figure 10: Attacker ping of the Web Server, Command

All pings were blocked by the firewall, which was reflected in the firewall log.

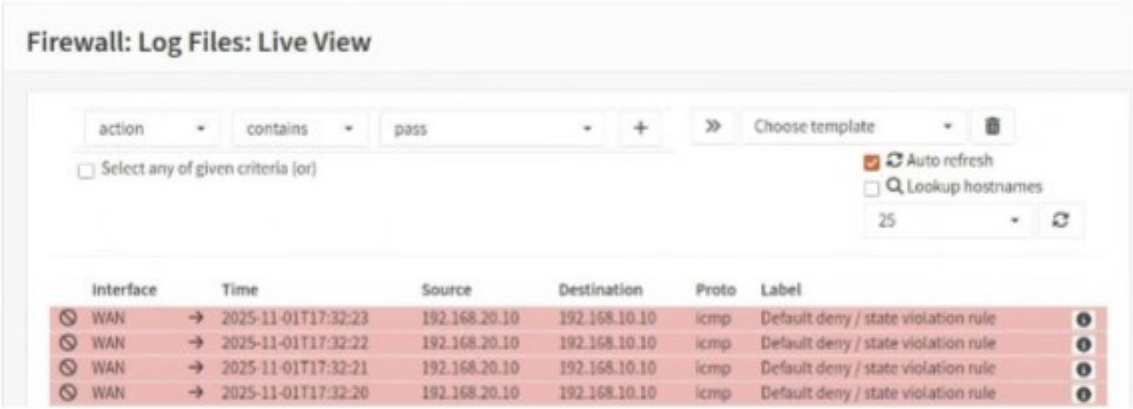


Figure 11: Attacker ping of the Web Server, Firewall Log

There were no intrusion detection alerts.

Port Scanning

As the Attacker PC is on the WAN network, it's only allowed to connect to ports 80 and 443 on the Web Server for HTTP and HTTPS. The firewall is blocking all the other connections.

SYN Scan

SYN scan had detected an HTTP service running on port 80 of the Web Server.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sS -Pn 192.168.10.10

Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 17:47 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid
servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0056s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 6.83 seconds
```

Figure 12: SYN Scan of Web Server, Command

The firewall log shows that all packets were blocked except those to port 80 (only a snippet of the log shown).

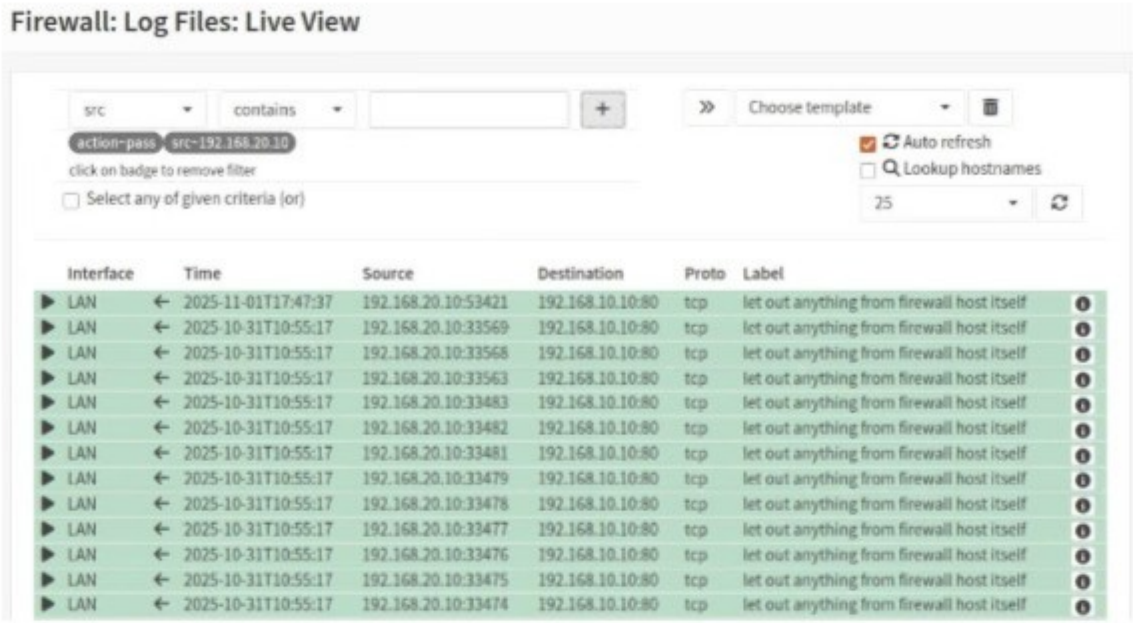


Figure 13: SYN Scan of Web Server, Firewall Log (passed)

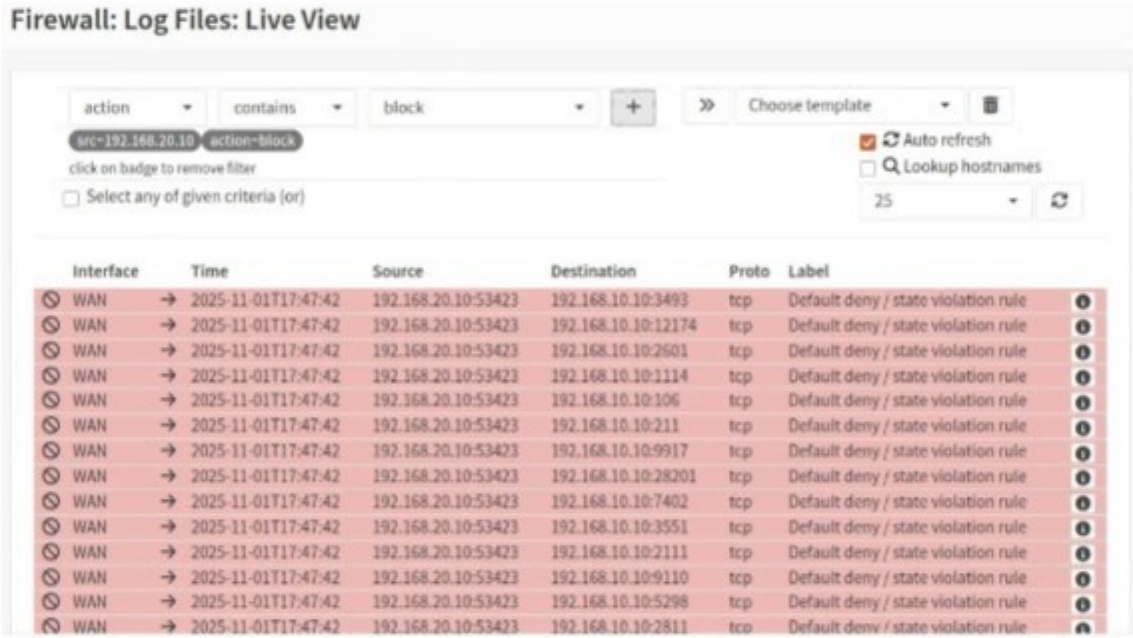


Figure 14: SYN Scan of Web Server, Firewall Log (blocked)

There were no intrusion detection alerts.

Connect Scan

SYN scan had detected an HTTP service running on port 80 of the Web Server.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sT -Pn 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 18:09 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid
servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0055s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 6.61 seconds
```

Figure 15: Connect Scan of Web Server, Command

The firewall log shows that all packets were blocked except those to port 80 (only a snippet of the log shown).

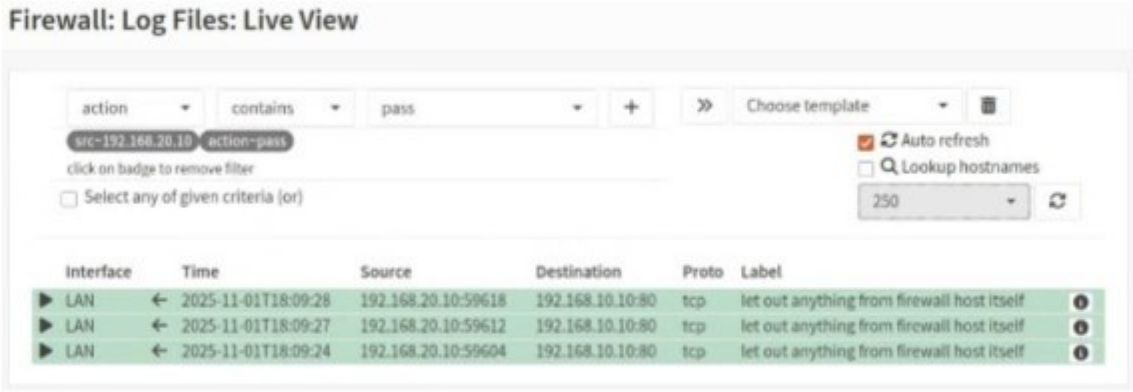


Figure 16: Connect Scan of Web Server, Firewall Log (passed)

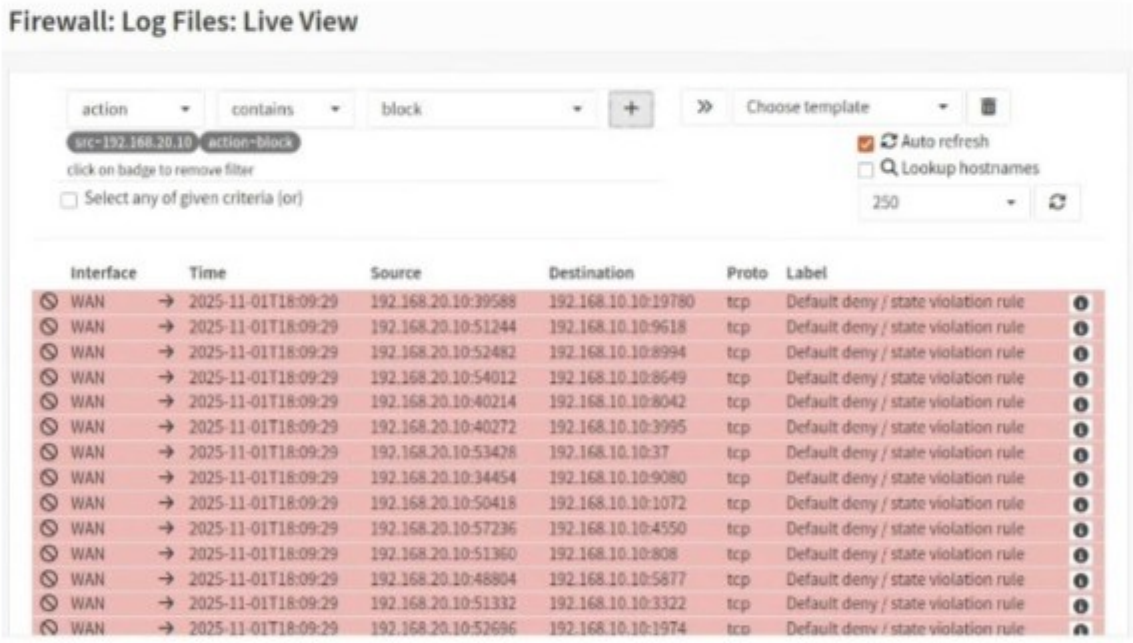


Figure 17: Connect Scan of Web Server, Firewall Log (blocked)

There were no intrusion detection alerts.

UDP Scan

The UDP scan has failed.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sU -Pn 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 18:20 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid
servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up.
All 1000 scanned ports on 192.168.10.10 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 205.96 seconds
```

Figure 18: UDP Scan of Web Server, Command

The firewall log shows that all packets were blocked (only a snippet of the log shown).

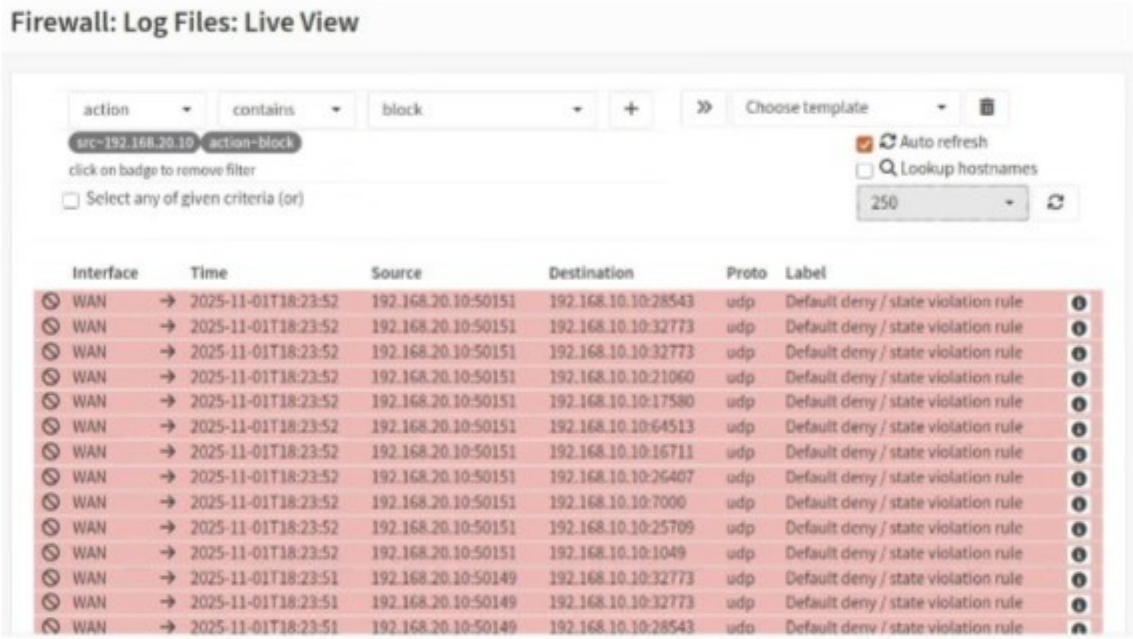


Figure 19: UDP Scan of Web Server, Firewall Log

There were no intrusion detection alerts.

Vulnerability Scanning

Vulnerability scanning found that the Web Server was running an HTTP service on port 80. Additionally, it showed that the Web Server is a Linux Ubuntu system running nginx version 1.24.0.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sV -Pn 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 18:33 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid
servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0035s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.24.0 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.18 seconds
```

Figure 20: Non-Script Vulnerability Scanning of Web Server, Command

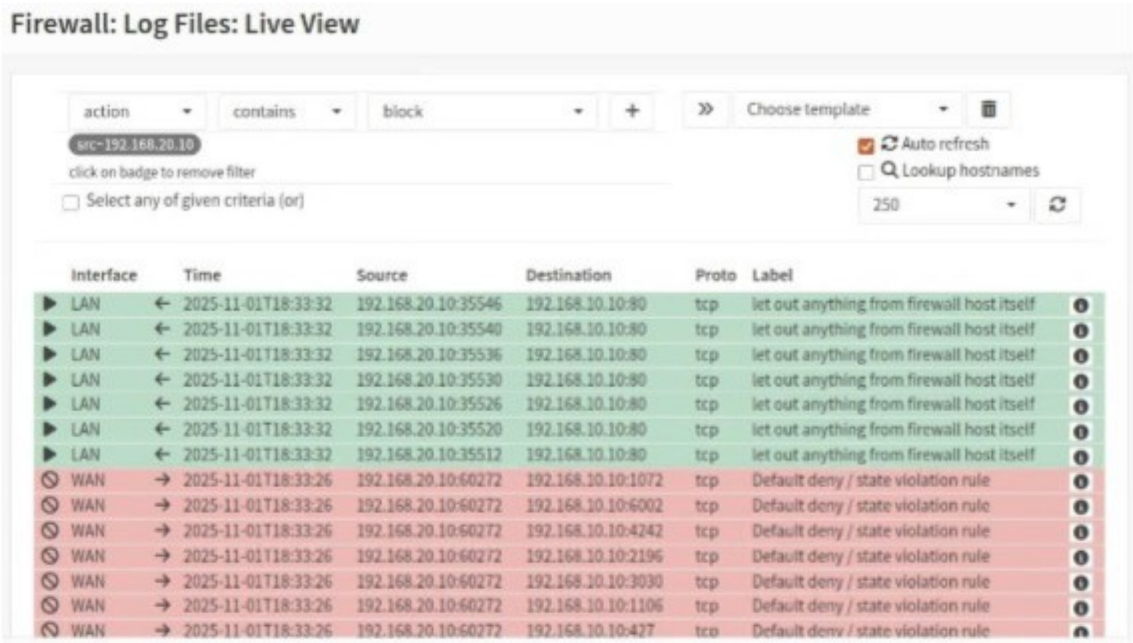


Figure 21: Non-Script Vulnerability Scanning of Web Server, Firewall Log

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sV -Pn --script vuln 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 18:39 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid
servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0045s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.24.0 (Ubuntu)
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-server-header: nginx/1.24.0 (Ubuntu)
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 101.38 seconds
```

Figure 22: Script Vulnerability Scanning of Web Server, Command

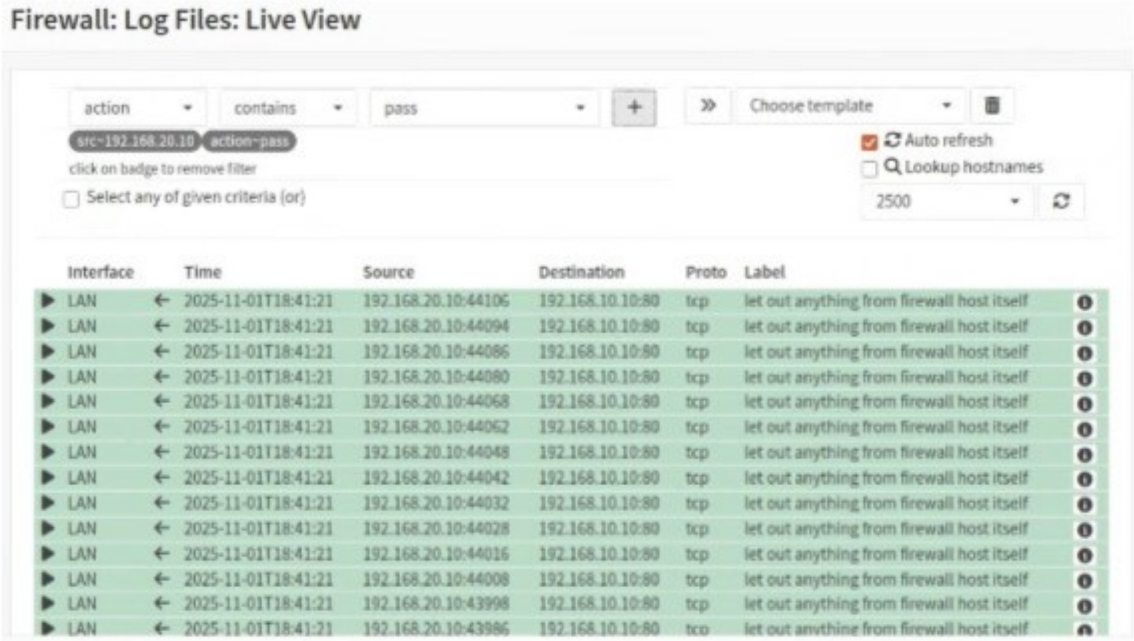


Figure 23: Script Vulnerability Scanning of Web Server, Firewall Log (passed)

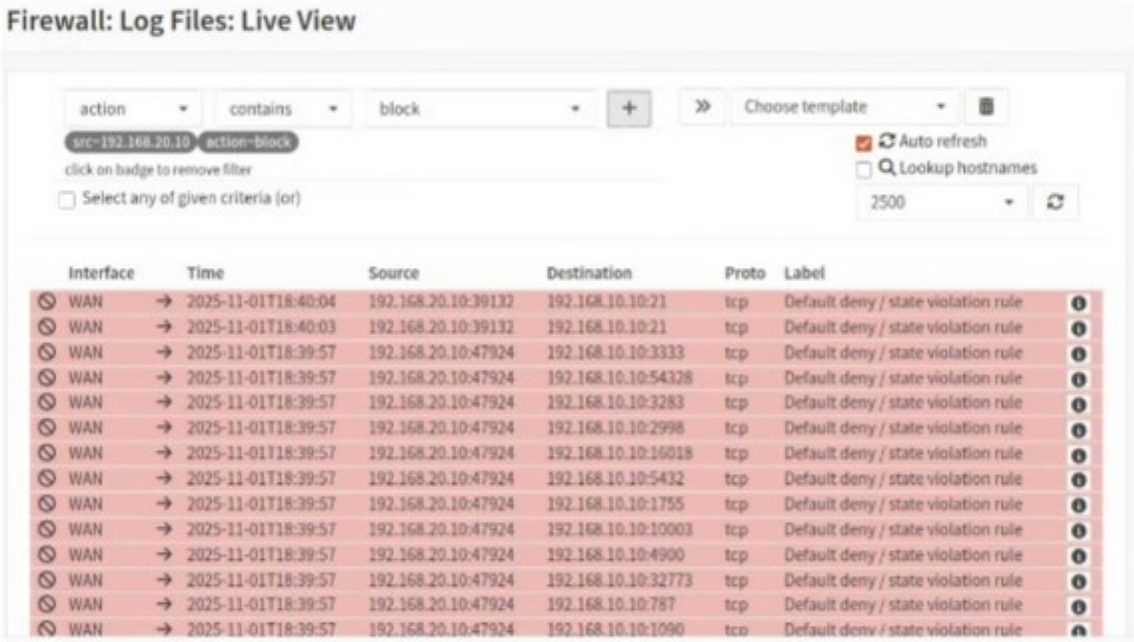


Figure 24: Script Vulnerability Scanning of Web Server, Firewall Log (blocked)

The intrusion detection service has sent alerts of a suspected Nmap scan.

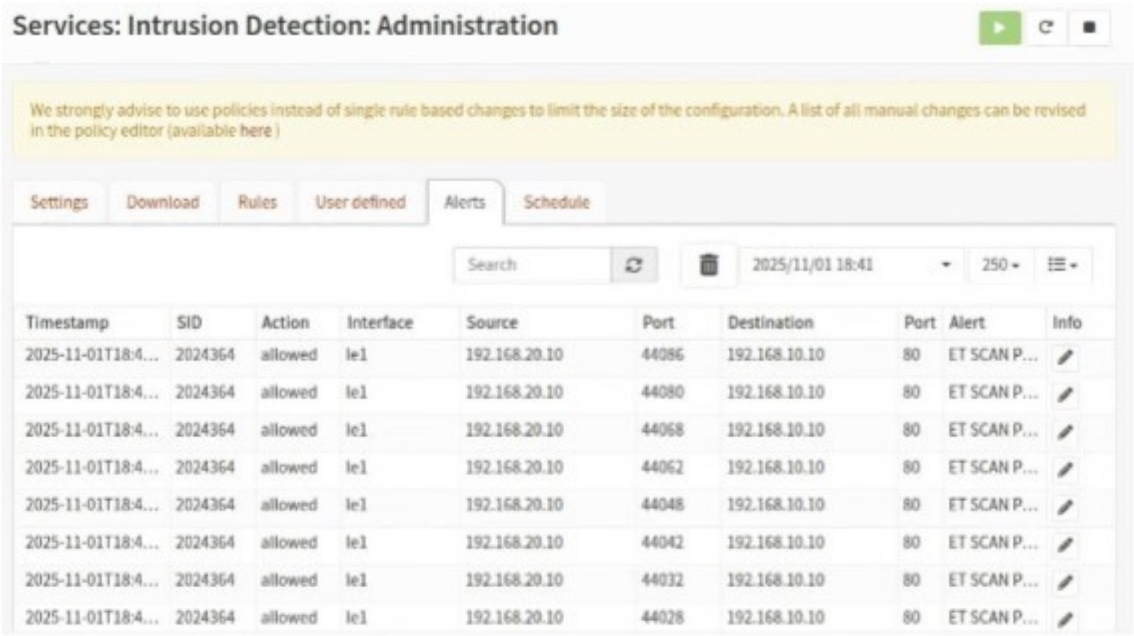


Figure 25: Script Vulnerability Scanning of Web Server, Intrusion Detection Alerts

Brute Force / Credential Attacks

The brute-force attack was unsuccessful because all SSH connections from the WAN that are not from the Admin PC are blocked by the firewall.



Figure 26: passwords.txt Contents

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hydra -l admin -P passwords.txt ssh://192.168.10.10:22
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, o
r for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-01 18:57:48
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 30 login tries (l:1/p:30), ~2 tries per task
[DATA] attacking ssh://192.168.10.10:22/
[ERROR] could not connect to ssh://192.168.10.10:22 - Timeout connecting to 192.168.10.10
```

Figure 27: SSH Brute Force Attack on Web Server, Command

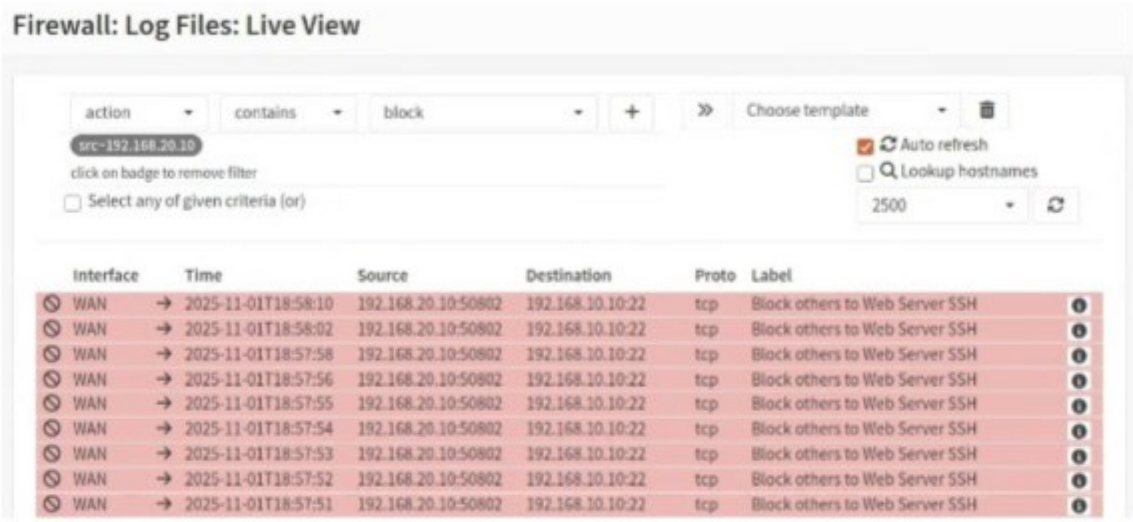


Figure 28: SSH Brute Force Attack on Web Server, Firewall Log

Denial of Service Attack

I have done DoS attack on the firewall WAN interface port 80 which has lead to GUI breaking.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ sudo hping3 -S -p 80 --flood 192.168.20.1
HPING 192.168.20.1 (eth0 192.168.20.1): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.20.1 hping statistic ---
1615884 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

┌──(attackerpc㉿attackerpc)-[~]
└─$
```

Figure 29: DoS Attack on Firewall WAN port, Command

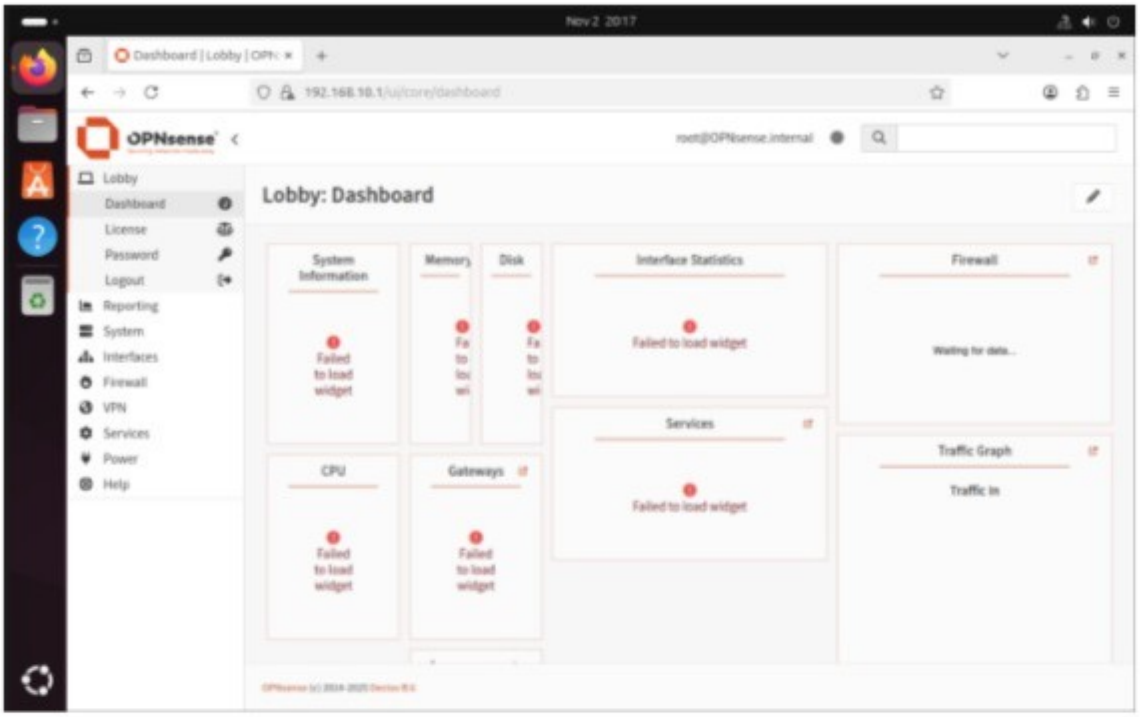


Figure 30: DoS Attack on Firewall WAN port, OPNsense Broken GUI

The firewall didn't block the packets coming to port 80, because it was exposed for port forwarding.

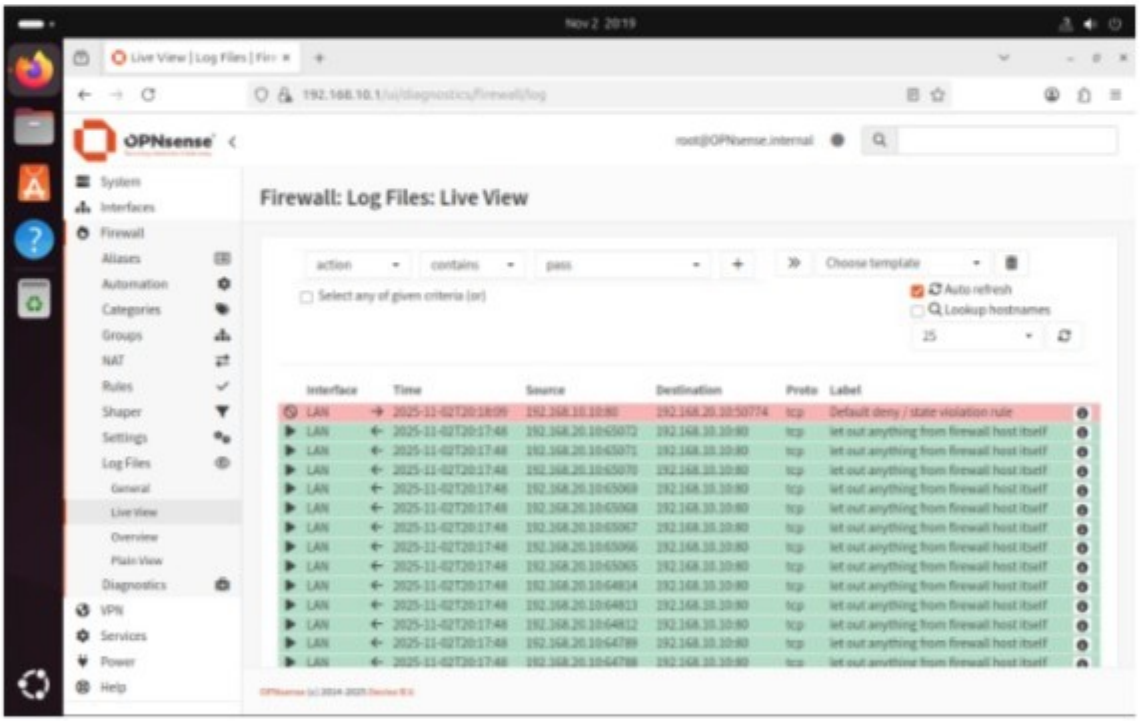


Figure 31: DoS Attack on Firewall WAN port, Firewall Log

Conclusion

OPNsense is a great firewall. I have enjoyed working with it. It was easy to set firewall rules. The live log was very useful for observing attacks and normal traffic. Although the intrusion detection system was overwhelming. Each ruleset has 100–120 rules, and you are not

supposed to enable all of them. Choosing which IDS rules you need is not straightforward. That is partially why many attacks I tested weren't detected by the IDS: I didn't select all the rules I needed. If I were to do a similar project in the future, I would use aliases to group admin users for easier rule configuration.

Smoothwall – Maël Coneios Crusson – [redacted]

Introduction

Smoothwall is a simple, easy to use, basic firewall. It is web-based, with a simpler, older style. It has key features such as Stateful Packet Inspection (SPI), basic filtering, and a simple VPN.

Firewall's Default Behaviour

The default behaviour for this firewall is to have no custom rules set. I will test the selected attacks to examine the firewall's default response.

Reconnaissance

Pinging the WAN_NET interface on the firewall (also known as 'Orange' interface or 'eth0' interface) was unsuccessful:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ ping -c 4 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
From 192.168.20.1 icmp_seq=1 Destination Port Unreachable
From 192.168.20.1 icmp_seq=2 Destination Port Unreachable
From 192.168.20.1 icmp_seq=3 Destination Port Unreachable
From 192.168.20.1 icmp_seq=4 Destination Port Unreachable

--- 192.168.20.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3174ms
```

Figure 29: Attacker PC pinging Smoothwall's WAN_NET Interface

Port Scanning

Using Nmap, I targeted the Web Server using a standard TCP SYN scan as well as a stealth scans. They were also unsuccessful:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sS 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 12:12 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.14 seconds

┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sF 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 12:12 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.12 seconds

┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sX 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 12:13 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.15 seconds

┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sN 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 12:13 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.14 seconds
```

Figure 30: Attacker PC's Port Scanning on Web Server

Vulnerability Scanning

Using the Nmap Scripting Engine was unsuccessful because HTTP (port 80) has not been allowed from the WAN_NET network to the Web Server:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sV --script vuln 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-26 12:24 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 13.66 seconds
```

Figure 31: Attacker PC's Vulnerability Scanning on Web Server

Brute-Force Attack

Using hydra was also unsuccessful because the attempted usernames were not the correct ones and SSH was not allowed from the WAN_NET network to the Web Server:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hydra -l admin -P passwords.txt ssh://192.168.10.10:22
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-26 13:05:
25
[WARNING] Many SSH configurations limit the number of parallel tasks, it is reco
mmended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 104 login tries (l:1/p:104),
 ~7 tries per task
[DATA] attacking ssh://192.168.10.10:22/
[ERROR] could not connect to ssh://192.168.10.10:22 - Connection refused
```

Figure 32: Attacker PC's Brute-Force on Web Server

Part of the passwords.txt file:

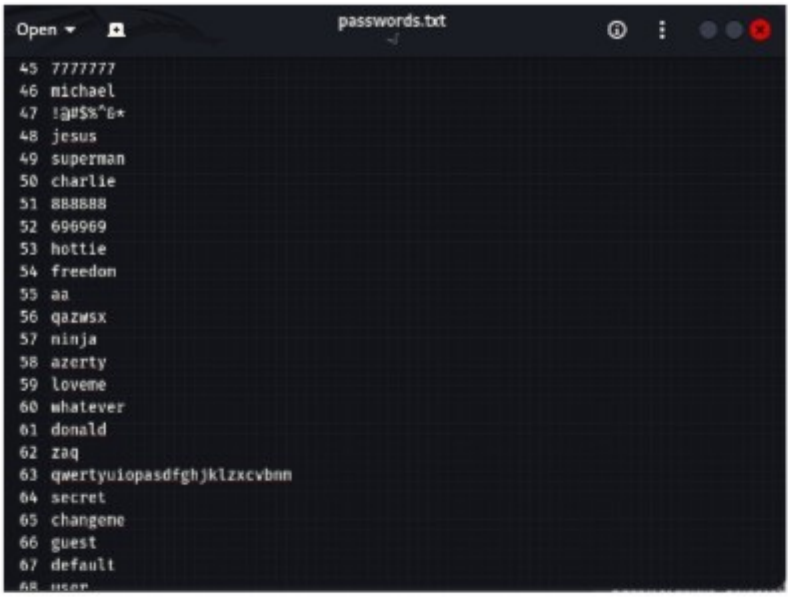


Figure 33: Part of passwords.txt's file content

Denial-of-Service

The three Hping attacks were also unsuccessful because opening a raw socket requires administrator privileges on Linux systems, which I did not know. Ultimately it's because HTTP (port 80) traffic was not allowed from the WAN_NET network to the Web Server:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hping3 -S -p 80 -c 10000 192.168.10.10
[open_sockraw] socket(): Operation not permitted
[main] can't open raw socket

┌──(attackerpc㉿attackerpc)-[~]
└─$ hping3 --flood 192.168.10.10
[open_sockraw] socket(): Operation not permitted
[main] can't open raw socket

┌──(attackerpc㉿attackerpc)-[~]
└─$ hping3 --flood 192.168.20.1
[open_sockraw] socket(): Operation not permitted
[main] can't open raw socket
```

Figure 34: Attacker PC's Dos attempt on Webserver

Conclusion for Smoothwall's Default Behaviour

It appears that the default configuration of Smoothwall Express is enough to prevent the attempted attacks, since with no specific rules set, it blocks absolutely everything from the WAN_NET network.

Setup & Configuration

From the Home PC we can access the firewall's GUI (https://192.168.10.1:441). I have come across issues with it because Smoothwall Express' web interface is designed for older browsers. Therefore, all commands/rules will be added to the firewall's startup file 'rc.firewall.up'.

Allow Admin to access LAN_NET

Since Admin PC is a trusted device in the WAN_NET network, it should have full access to LAN_NET's network:

```
# Custom Rules
#   Adding FORWARD rule: Admin PC to LAN_NET
iptables -A FORWARD -i eth0 -o eth1 -s 192.168.20.20 -d 192.168.10.0/24 -j ACCEP
T
```

Figure 35: Added rule for Admin PC on Smoothwall

Enabling ICMP

For the reconnaissance attack to work, ICMP pings should be enabled. In the GUI, the option for blocking ICMP ping is disabled but the firewall's configuration still prevents ICMP ping.

I removed the two rules that blocked all ICMP pings returning 'Destination Port Unreachable' on the Attacker PC. I searched for the rules blocking all ICMP pings returning 'Destination Port Unreachable' using the commands:

```
iptables -L INPUT -n -v --line-numbers
iptables -L FORWARD -n -v --line-numbers
```

Figure 36: Commands to find the blocking rules

I found the two rules to be deleted:

- '17 1 2139 REJECT all -- * * 0.0.0.0/0 0.0.0.0/0 reject-with icmp-port-unreachable'
- '26 0 0 REJECT all -- * * 0.0.0.0/0 0.0.0.0/0 reject-with icmp-port-unreachable'

To remove the rules, these commands must be added:

```
#   Removing INPUT rule: REJECT ALL
iptables -D INPUT 17
#   Removing FORWARD rule: REJECT ALL
iptables -D FORWARD 26
```

Figure 37: Added rules for ICMP traffic on Smoothwall

A new rule must be added to accept all ICMP traffic to the firewall:

```
#   Adding INPUT rule: ALLOW icmp ALL
iptables -A INPUT -p icmp -j ACCEPT
```

Figure 38: Added rule for ICMP traffic on Smoothwall

HTTP and HTTPS

To allow the vulnerability scanning and denial-of-service attacks to work, traffic must be allowed on (for example) port 80 (HTTP) from the WAN_NET network to the Web Server. I will also be allowing traffic on port 443 (HTTPS).

On the web interface, adding the rules to incoming ports was actually possible:

Current rules:

Protocol	External source IP	Original destination port or range	New destination IP	New destination port or range	Enabled	Mark
Comment						
TCP	192.168.20.0/24	HTTP (80)	192.168.10.10	HTTP (80)	✓	☐
TCP	192.168.20.0/24	HTTPS (443)	192.168.10.10	HTTPS (443)	✓	☐

Remove Edit

Figure 39: Added HTTP and HTTPS rules on Smoothwall's Web Interface

Configuration can also be made in the CLI:

```
#   Adding FORWARD rule: HTTP traffic
iptables -A FORWARD -i eth0 -o eth1 -s 192.168.20.0/24 -d 192.168.10.10 -p tcp -
-dport 80 -j ACCEPT
#   Adding FORWARD rule: HTTPS traffic
iptables -A FORWARD -i eth0 -o eth1 -s 192.168.20.0/24 -d 192.168.10.10 -p tcp -
-dport 443 -j ACCEPT
```

Figure 40: Added HTTP and HTTPS rules on Smoothwall's CLI

SSH

For the brute-force attack to succeed, SSH (port 22) must be allowed from the WAN_NET network to the Web Server:

```
#   Adding FORWARD rule: ALLOW SSH
iptables -A FORWARD -i eth0 -o eth1 -s 192.168.20.0/24 -d 192.168.10.10 -p tcp -
-dport 22 -j ACCEPT
```

Figure 41: Added rule for SSH traffic on Smoothwall

Attacks

Reconnaissance (Active)

Pinging the WAN_NET interface on the firewall is now successful:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=0.906 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=64 time=1.00 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=64 time=1.17 ms
^C
--- 192.168.20.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3016ms
rtt min/avg/max/mdev = 0.906/1.033/1.174/0.096 ms
```

Figure 42: Attacker PC pings Smoothwall's WAN_NET Interface

Pinging the Web Server remains correctly unsuccessful:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
From 192.168.20.1 icmp_seq=1 Destination Port Unreachable
From 192.168.20.1 icmp_seq=2 Destination Port Unreachable
From 192.168.20.1 icmp_seq=3 Destination Port Unreachable
From 192.168.20.1 icmp_seq=4 Destination Port Unreachable
^C
--- 192.168.10.10 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3009ms
```

Figure 43: Attacker PC pings Web Server

Using Nmap scan was successful, identifying all devices in WAN_NET: firewall, Admin PC and Attacker PC:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sn 192.168.20.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 10:17 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.20.1
Host is up (0.0013s latency).
MAC Address: 08:00:27:35:A1:AC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.20.20
Host is up (0.00072s latency).
MAC Address: 08:00:27:C9:DE:DA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.20.10
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.07 seconds
```

Figure 44: Attacker PC Nmap scans WAN_NET Network

Port Scanning

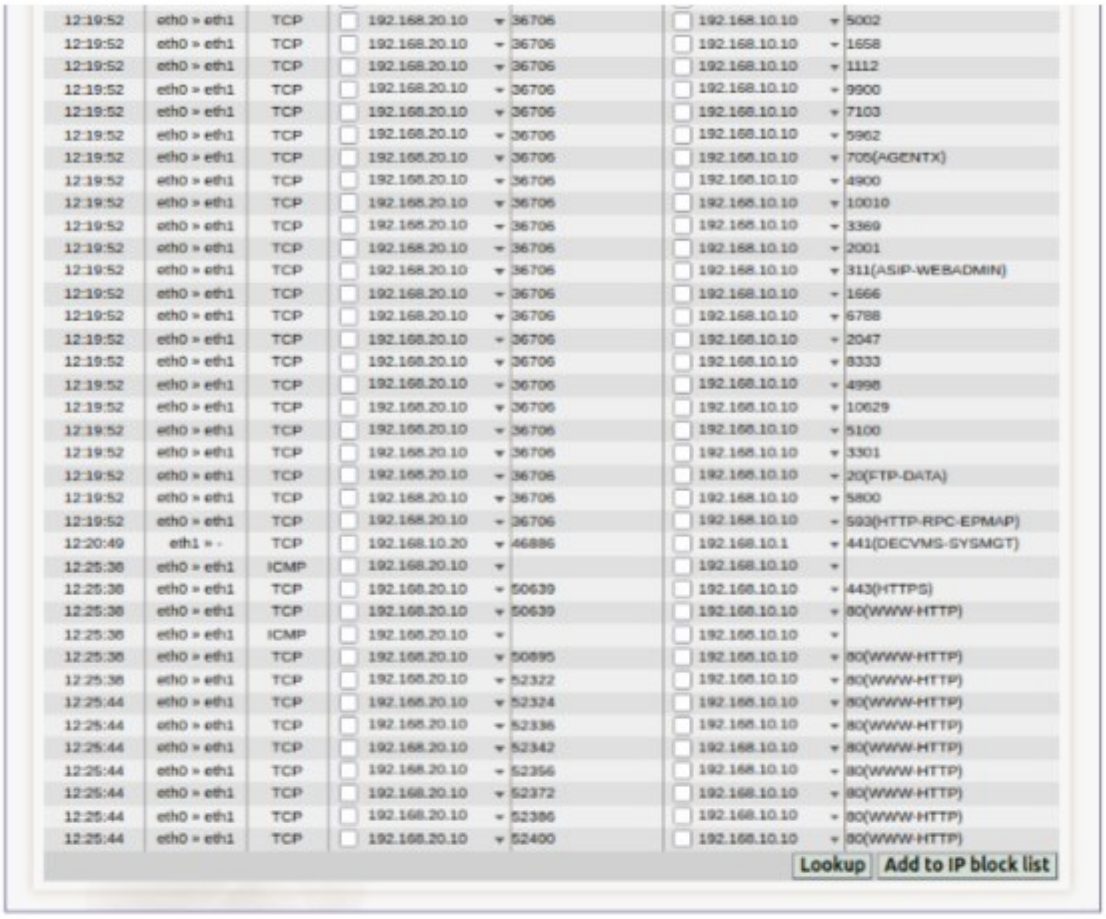
The expected result from the port scanning is that port 80 alone will be forwarded, which was a success:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sS 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 11:19 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0029s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE  SERVICE
80/tcp  open   http
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 4.73 seconds
```

Figure 45: Attacker PC Port Scans Web Server

The logs demonstrate the port scanning:



12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	5002
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	1658
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	1112
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	9900
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	7103
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	5962
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	705(AGENTX)
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	4900
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	10010
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	3369
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	2001
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	311(ASIP-WEBADMIN)
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	1666
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	6788
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	2047
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	8333
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	4998
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	10629
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	5100
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	3301
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	20(FTP-DATA)
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	5800
12:19:52	eth0 » eth1	TCP	192.168.20.10	36706	192.168.10.10	593(HTTP-RPC-EPMAP)
12:20:49	eth1 » -	TCP	192.168.10.20	46886	192.168.10.1	441(DECVMS-SYSMGT)
12:25:36	eth0 » eth1	ICMP	192.168.20.10		192.168.10.10	
12:25:36	eth0 » eth1	TCP	192.168.20.10	50639	192.168.10.10	443(HTTPS)
12:25:36	eth0 » eth1	TCP	192.168.20.10	50639	192.168.10.10	80(WWW-HTTP)
12:25:36	eth0 » eth1	ICMP	192.168.20.10		192.168.10.10	
12:25:36	eth0 » eth1	TCP	192.168.20.10	50895	192.168.10.10	80(WWW-HTTP)
12:25:36	eth0 » eth1	TCP	192.168.20.10	52322	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52324	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52336	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52342	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52356	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52372	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52386	192.168.10.10	80(WWW-HTTP)
12:25:44	eth0 » eth1	TCP	192.168.20.10	52400	192.168.10.10	80(WWW-HTTP)

Lookup | Add to IP block list

Figure 46: Attacker PC's Port Scanning on Smoothwall Log

Vulnerability Scanning

Nmap scans are a success, showing the version of the web server: 'nginx 1.24.0 (Ubuntu)':

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sV 192.168.10.10 -p 80
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 11:56 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0037s latency).

PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.24.0 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.69 seconds
```

Figure 47: Attacker PC Nmap Scans Web Server

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sV --script vuln 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-29 14:09 GMT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
 Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0026s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE  SERVICE VERSION
22/tcp  open   ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol
 2.0)
80/tcp  open   http    nginx 1.24.0 (Ubuntu)
|_http-server-header: nginx/1.24.0 (Ubuntu)
| http-vuln-cve2011-3192:
|   VULNERABLE:
|   Apache byterange filter DoS
|     State: VULNERABLE
|     IDs:  CVE:CVE-2011-3192  BID:49303
|       The Apache web server is vulnerable to a denial of service attack when n
umerous
|       overlapping byte ranges are requested.
|     Disclosure date: 2011-08-19
|     References:
|       https://www.tenable.com/plugins/nessus/55976
|       https://www.securityfocus.com/bid/49303
|       https://seclists.org/fulldisclosure/2011/Aug/175
|_      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-3192
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
443/tcp closed https
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.59 seconds
```

Figure 48: Attacker PC Nmap Scans Web Server

This demonstrates that the firewall is a border guard, not a vulnerability patcher. Once a rule allows traffic, the service itself must be secure.

Brute-Force Attack

Since I put the correct password in the passwords.txt file and put the right username, it was a success:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hydra -l webserver -P passwords.txt 192.168.10.10 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-29 12:44:51
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce
the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 104 login tries (l:1/p:104), ~7 tries per task
[DATA] attacking ssh://192.168.10.10:22/
[22][ssh] host: 192.168.10.10   login: webserver   password: Pass1234
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 1 final worker threads did not complete until end.
[ERROR] 1 target did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-29 12:45:14
```

Figure 49: Attacker PC's Brute-Force on Web Server

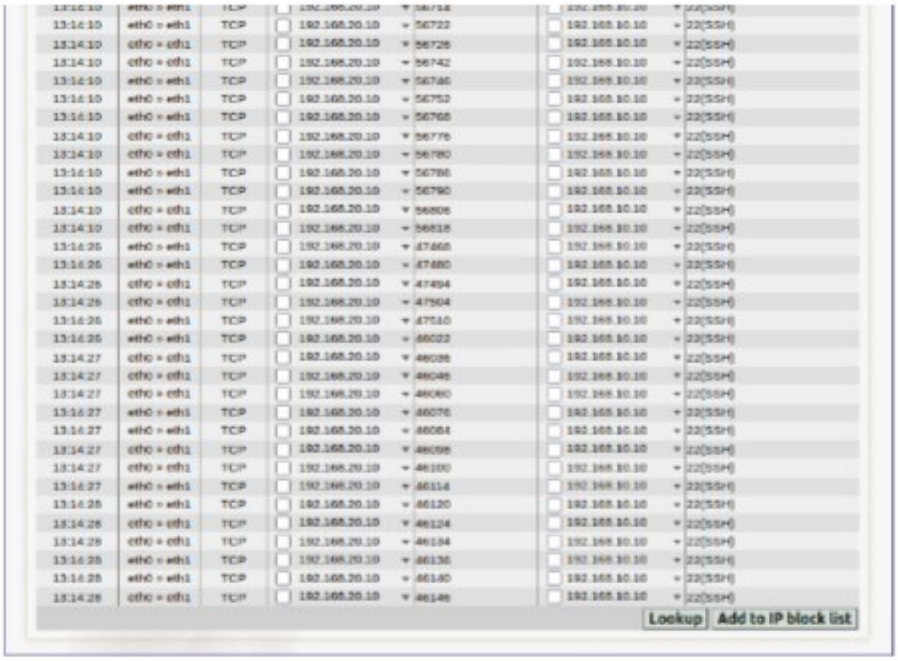


Figure 50: Attacker PC's Brute-Force on Smoothwall's Log

This shows that the firewall passes the traffic until the Web Server's SSH service blocks the attempts. The firewall will not block by default.

Denial-of-Service

With administrator privileges and the added rule, the attack was a success:

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ sudo hping3 -S -p 80 --flood 192.168.10.10
[sudo] password for attackerpc:
HPING 192.168.10.10 (eth0 192.168.10.10): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Figure 51: Attacker PC's Dos on Web Server

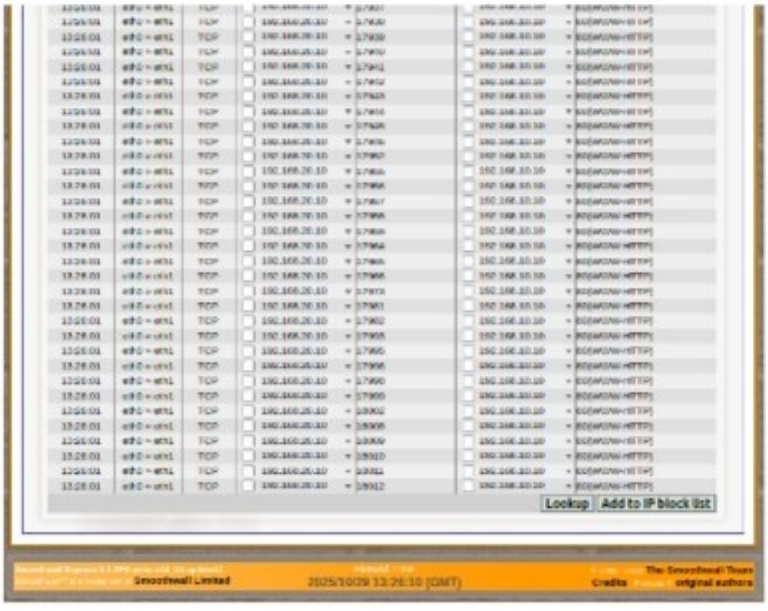


Figure 52: Attacker PC's Dos on Smoothwall's Log

Pinging from Admin PC before the attack, the packets return at normal speed (3003ms):

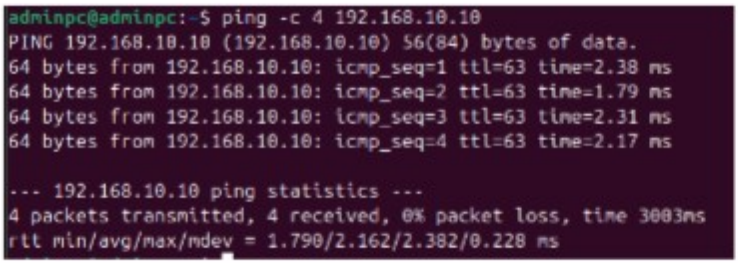


Figure 53: Ping from Admin PC to Web Server

When the attack is ongoing, the packets return at slower speed or even fail:

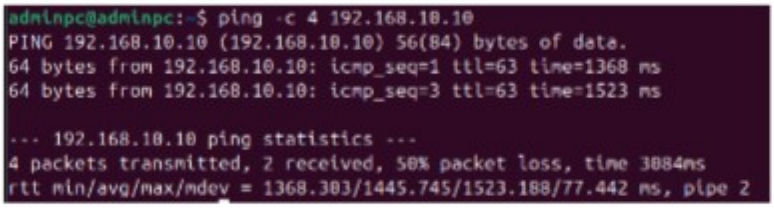


Figure 54: Ping from Admin PC to Web Server

Conclusion

To conclude, Smoothwall performs quite well as a defence, matching the desired network behaviour by blocking all unwanted inbound traffic from the WAN network by default. To meet the requirements, rules had to be added for Smoothwall to allow HTTP/HTTPS to the Web Server and for the Admin PC to access the Web Server via SSH. The success of the Vulnerability Scanning and Brute-Force Attack demonstrated that once a rule explicitly permits traffic, the firewall simply forwards it, showing its function as a border guard that does not patch vulnerabilities or secure the internal services. The successful Denial-of-Service Attack highlighted that even with strong default security, implementing non-default features like rate-limiting is necessary to prevent resource exhaustion and ensure the stability of the services exposed through port forwarding.

VyOS – Joseph Cronin [redacted]

Introduction

VyOS is an open source, Linux operating system that allows the user to create routers and firewalls for a wide range of uses. A Debian-based system, it uses solely the CLI as it's method of user interaction and there is no graphical interface. It is free to use, can be integrated into physical hardware, cloud environments and in the case of this project, as a virtual machine.

Configuration

VyOS required a permanent installation once the VM was created inside of Virtual Box. Prior to this, each time the IP addresses were configured to eth0 and eth1, and the machine was rebooted, the entire system was reverted back to a fresh install.
However, once all Virtual Machines were installed and configured with IP addresses, each machine was freely able to ping the other machines.
The screenshot below shows the AttackerPC pinging the Ubuntu WebServer as one example of this.

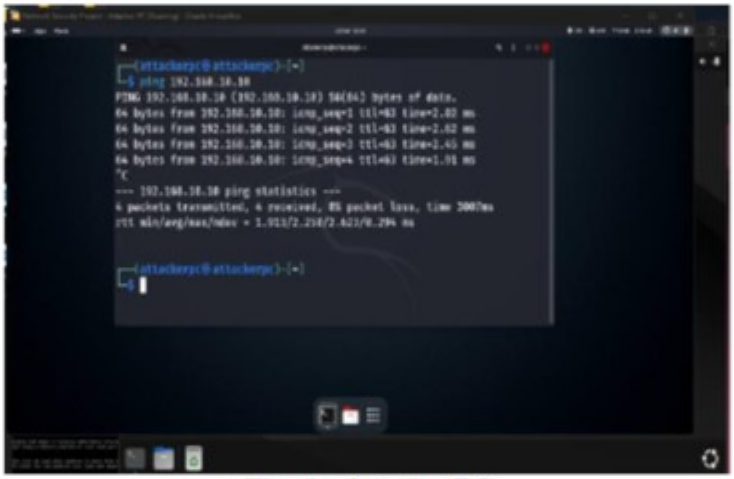


Fig. 1 - AttackerPC

The rest of the configuration for this project was performed within the VyOS system.

In the most recent versions of VyOS (1.4.1 onwards) Zone Based Firewalls were introduced. In short, this means that each interface is assigned a zone and each firewall is then also attached to the relevant zone.
As shown in the screenshot below, the zones LAN, WAN and LOCAL have been created and each zone has been initially configured to drop any traffic

```
vyos@vyos# run show firewall zone
Zone    Members    From Zone    Firewall IPv4    Firewall IPv6
------  ---------  -----------  ---------------  ---------------
LAN     eth0       WAN          WAN-IN
LOCAL   LOCAL      LAN          LAN-ROUTER
                   WAN          WAN-ROUTER
WAN     eth1       LAN          LAN-OUT
[edit]
vyos@vyos# _
```

Fig. 2 - Zone Based Firewalls

The LAN Zone has been assigned to the interface eth0.

The WAN Zone has been assigned to the interface eth1.

The LOCAL Zone is the router itself.

The firewall configurations themselves have varying levels of complexity, ranging from 2 to 4 rules (10, 20, 30, 40) in addition to the default drops.

WAN-IN controls the traffic from outside the External Network (WAN) to the Internal Network (LAN) while LAN-OUT controls the traffic going the opposite way.
The WAN-ROUTER firewall (for the purpose of the project) has been configured to allow the connection to the AdminPC, and drop the connection from the AttackerPC.
The LAN-ROUTER firewall has been configured to allow SSH from the HomePC and prevents the WebServer from accessing the router.

Screenshots of each firewall configuration are below.

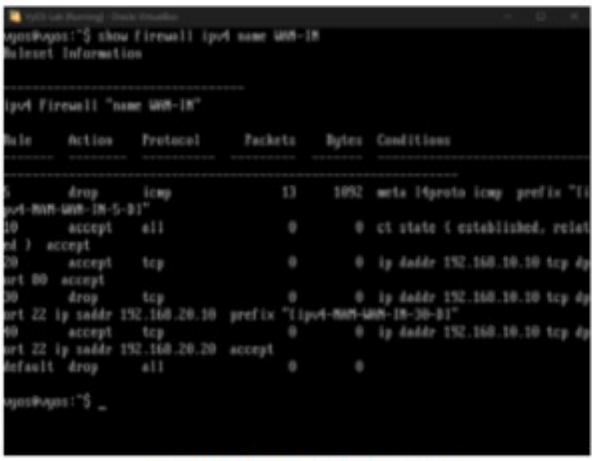


Fig 3. WAN-IN Firewall



Fig 4. LAN-OUT Firewall

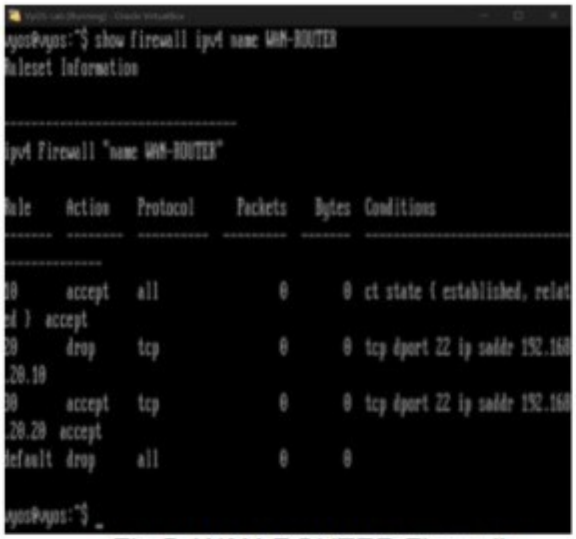


Fig 5. WAN-ROUTER Firewall

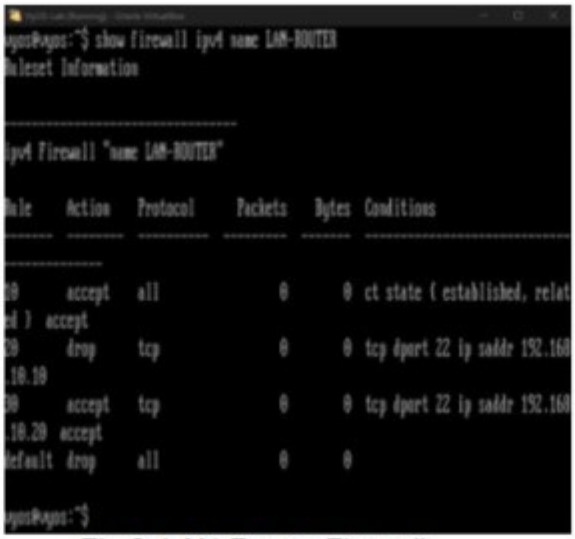


Fig 6. LAN-Router Firewall

Attacks

Active Reconnaissance / Ping

The first attack performed was in the form of a basic Ping. This ping demonstrates a very simple form of active reconnaissance on the Web Server from the AttackerPC.

From the point of view of the AttackerPC, there has been no information received from the ping. Two packers have been transmitted and 0 received.

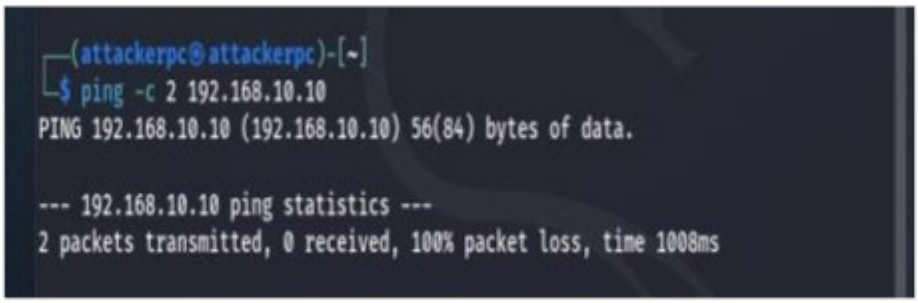


Fig. 7 - AttackerPC Ping to WebServer

Using the log functions within VyOS, it is possible to see the instances where the pings were received and then dropped because of the firewall configuration.

[ipv4-NAM-WAN-IN-5-D] shows that Rule 5 was triggered from within firewall WAN-IN, dropping the ping and sending no response back to the AttackerPC.

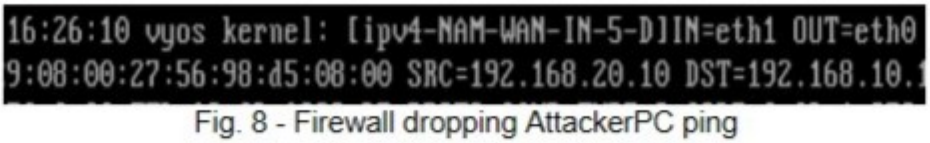


Fig. 8 - Firewall dropping AttackerPC ping

Active Reconnaissance / Port/Syn Scanning

The second attack that was conducted was Port/SYN scanning, where the AttackerPC can search for open ports on the Web Server host. The result of the scan shows that Port 80 is open for HTTP services. This allows for devices outside of the LAN environment to access a website that may be hosted on the webserver.



Fig. 9 – AttackerPC SynScan on WebServer

The log file in this instance shows [ipv4-NAM-WAN-IN-30-D] showing that rule 30 was triggered from within the WAN-IN firewall. This rule is configured (in this case specifically for AttackerPC) to prevent access to the SSH Service, dropping the connection.



Fig. 10 – Firewall dropping AttackerPC SynScan (Blocking SSH)

In contrast, the AdminPC is successfully able to access the WebServer as seen in the screenshots below.

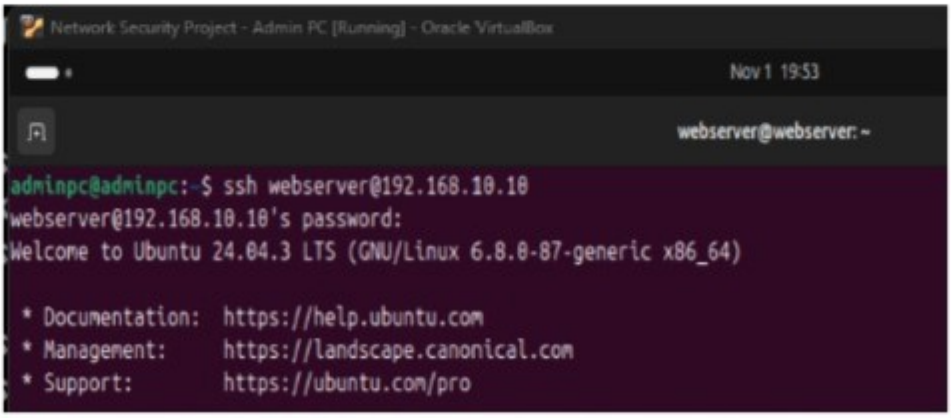


Fig. 11 – AdminPC successful SSH to WebServer

Brute Force

The current firewall configuration prevents any attempted connection from the AttackerPC to the Web Server. In order to attempt a Brute Force attack, the WAN-IN firewall must first be modified.
First disabling Rule 30, which specifies that the AttackerPC IP Address is dropped at any attempted connection (This is visible in Fig. 1 above) and then adding an additional rule to allow the AttackerPC through the firewall.

```
35      accept    tcp              17     1020  tcp dport 22 ip saddr 192.168
.20.10  accept
```

Fig. 12 – Rule 35 addition to WAN-IN Firewall

Once the modifications were made to the firewall, a Brute Force attack could be attempted using the rockyou.txt file within the Linux Kali OS. With over 14 million passwords to try, completing the attack at this rate would take months.

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hydra -l webserver -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.10 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations
, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-01 20:47:41
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.10.10:22/
[STATUS] 64.00 tries/min, 64 tries in 00:01h, 14344335 to do in 3735:31h, 4 active
[STATUS] 68.00 tries/min, 204 tries in 00:03h, 14344195 to do in 3515:45h, 4 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

Fig. 13 – Successful connection to WebServer from AttackPC / Hydra Brute Force Attack

Conclusion

Originally, a stateful firewall was created within VyOS in order to control the network traffic between the different hosts. However, it did not seem possible to apply this firewall to interfaces eth0 and eth1.
Zone-based firewalls resolved this issue as each rule/firewall created could be assigned to a particular zone, and each zone then applied to either of the interfaces, or the router itself.

The Command Line Interface can be quite challenging, compared to a visual GUI, and it took some time get completely used to working in solely in that environment. VyOS stands out as quite strict, and obviously, will only work if each command is correct and complete but overall VyOS is a powerful router and firewall.

Bibliography

Joseph - VyOS

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